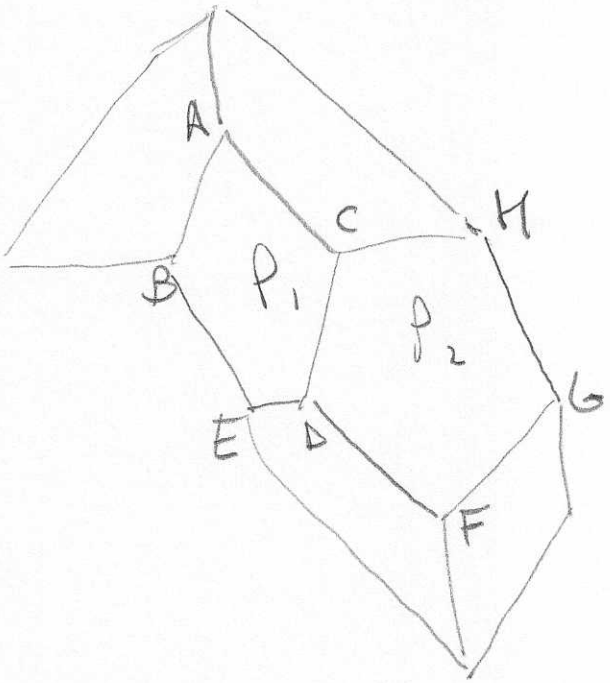
3273

The diagram is



Notice that AB and AC are congruent and so are DF and FG. In particular P_1 and P_2 are both "isosceles pentagons". Since these pentagons are determined by the length of any side, P_1 and P_2 are congruent.

So we also have that ED = CH

and in particular the quadrilaterals are congruent letting

$$b = \tan\frac{AC}{2} \quad \text{and} \quad c = \tan\frac{CH}{2} = \tan\frac{ED}{2}$$

we have the relations

$$c = \frac{1}{2b}$$

and $$c^2 = \frac{2b^4 + 10b^2 - 1}{(b^2-2)^2}$$

Substituting we find

$$(b^2-2)^2 = 4b^2(2b^4 + 10b^2 - 1)$$

namely $$8b^6 + 39b^4 - 4 = 0$$

This has only two real solutions and one of them is negative, while b is necessarily positive.

There is only one possible value for b, and hence for the entire configuration

This configuration is in fact a patron of Heppes 5

The derived curve net is then isometric to Heppes 7 in Figure 11 with the edge V_3V_7 removed.

This means that one of the plus has to be in the segment V_3V_7 and the other would need to be one of the vertices of the net, but this is not possible